Cø / N* Student Base Resource Pack

Key-free and reveal-safe collaborative materials for Sessions 1-14

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Student Use Guide

This pack contains initial student-facing materials only. It omits answer keys and every instructor-controlled reveal. Your instructor will release assigned evidence cards after the stated commitment point. All cases and values are synthetic and for teaching only; do not use them for clinical, welfare, or consciousness judgments outside the course.

Common Use Rules

Core and extension

- **Core** material is designed to fit the collaborative block in the session plan.
- **Extension** material may be assigned before class, used when a group finishes early, or moved to a discussion section.
- The student base pack contains neither keys nor controlled-reveal evidence. Instructors release only the assigned reveal sheet after the stated written commitment; answer keys remain private until the exercise closes.

Shared role scaling

Use the session-specific role labels where supplied. Scale them as follows.

Team size	Assignment rule
3	Student 1 combines model reconstruction and reporting. Student 2 is validity and methods auditor. Student 3 is rival advocate and process monitor.
4	Use model reconstructer, validity auditor, rival advocate, and synthesis reporter as separate roles.
5	Use the four separate roles and add a process/accessibility monitor who tracks target drift, time, dissent, and equitable participation.

Every student makes a private initial commitment before group discussion. Every team submits one conclusion, one residual uncertainty, and any minority note.

Accessible and digital default

All cards may be distributed as numbered LMS items or shared-document sections instead of paper. Students may respond in writing rather than move to a physical corner or speak in a fishbowl. Read every formula aloud in plain language. Do not encode status by color alone. Provide tables as editable text, allow screen-reader users to inspect one row at a time, and give students two quiet minutes before oral reporting. A student may serve as written analyst or asynchronous reviewer without disclosing a disability or personal experience.

Default numerical decision rules

Use these only when a session does not declare a different rule.

- A gate is **positive** when its lower uncertainty bound meets or exceeds its frozen threshold.
- A gate is **negative** when its upper uncertainty bound falls below its frozen threshold.
- A gate is **indeterminate** when its interval straddles the threshold.
- Equality is deliberate: a lower bound exactly on the threshold counts as a positive pass, whereas an upper bound exactly on the threshold does **not** count as a negative. If neither rule applies, report indeterminate.

- An equivalence interval passes only when the complete interval is contained in the declared **closed** margin, including its endpoints. A point estimate inside the margin or a nonsignificant difference is insufficient.
- A result is **unlicensed** when boundary, applicability, quality, source fidelity, independence, or transport fails.
- Credible surrogate conflict has precedence over scalar aggregation.
- A negative gate is a model-level negative prediction under the biconditional, not a direct observation of absent experience.

Session 1 Resource. Target Sorting and Inference Control

Student Materials

Core. Sort Cards 1-6. **Extension.** Add Cards 7-8 and one student-authored mixed-target card.

Target columns: wakefulness; minimal phenomenal presence C_0; content presence; access or task use; reflective or metacognitive consciousness C_1; phenomenal character.

For each card record: most direct target, possible relevance to C_0, required bridge, strongest alternative explanation, strongest licensed conclusion, and one forbidden stronger inference.

1. **Synthetic Card 1 - covert command following.** A behaviorally unresponsive patient modulates imagery-related activity on two preregistered command trials. Hearing, comprehension, and artifact checks pass.
2. **Synthetic Card 2 - early decoding.** Stimulus identity is decoded from early visual activity at 80 ms. No perturbation or downstream-use test was performed.
3. **Synthetic Card 3 - pain report.** A participant says, “The pain is sharp, unpleasant, and spreading.”
4. **Synthetic Card 4 - recurrent loop.** A content-specific recurrent loop persists for 250 ms in an isolated sensory module.
5. **Synthetic Card 5 - uncertainty model.** An artificial agent represents and reports calibrated uncertainty about its own decisions.
6. **Synthetic Card 6 - wakeful nonresponse.** A patient has sustained eye opening and sleep-wake cycles but produces no reliable command response.
7. **Synthetic Card 7 - blindsight-like choice.** A participant denies seeing a target but chooses its location at 78 percent accuracy under forced choice.
8. **Synthetic Card 8 - affective withdrawal.** A nonverbal animal withdraws selectively from one noxious stimulus and later avoids its location; sensory and motor controls pass. Use one row per assigned card:

Card	Most direct target	Possible relevance to C_0	Required bridge	Strongest alternative	Licensed conclusion	Forbidden stronger inference

Core deliverable: one six-row grid and a two-sentence rule distinguishing evidence from constitution.

Extension deliverable: append three rows for Cards 7-8 and the student-authored card. On the invented-card row, mark every implicated target before selecting its most direct target.

Accessible/digital alternative: use a shared grid with one card per row. Students enter private placements in hidden columns before the group column is opened.

Session 2 Resource. Component-Matrix Jigsaw

Frozen Rules

- Candidate reliability must satisfy $R_C \geq .80$ before a system is used as a P_Θ referent. A positive-bearer output additionally requires $R_B \geq .80$ plus stable mappings and the declared minimality / separability rules.
- V threshold: $.65$.
- N_1 threshold: $.60$.
- N_2 threshold: $.60$.
- N_3 threshold: $.50$.
- Two credible surrogates conflict when their intervals do not overlap and their point estimates differ by more than $.15$.

All intervals are 95 percent teaching intervals. All profiles are synthetic.

Student Materials

Core. Assign one profile per team. **Extension.** Teams classify a second profile and write a threat to the opposite direction of the biconditional.

Profile	Referent / bearer status	V	N_1	N_2	N_3	Additional information
A	$R_C = .92$; no all-pass proper subsystem; $R_B = .88$; positive bearer resolved	$.81[.74, .86]$	$.74[.68, .80]$	$.69[.63, .75]$	$.58[.52, .64]$	applicability, quality, mappings, and separability pass
B	stable P_Θ referent, $R_C = .90$; R_B not applicable to a positive output	$.79[.72, .84]$	$.73[.67, .79]$	$.41[.34, .48]$	$.61[.55, .67]$	content fidelity, report exclusion, and route tests pass
C	stable P_Θ referent, $R_C = .88$; bearer output unresolved	not measured	$.72[.65, .79]$	$.70[.64, .76]$	$.59[.53, .65]$	the preparation suffered an undocumented power interruption
D	stable P_Θ referent, $R_C = .91$; bearer output unresolved	$.80[.74, .85]$	A: $.72[.67, .77]$; B: $.47[.42, .52]$	$.68[.62, .74]$	$.57[.51, .63]$	both N_1 surrogates were preregistered and credible

Profile	Referent / bearer status	V	N_1	N_2	N_3	Additional information
E	candidate X $R_C = .74$; candidate Y $R_C = .76$	X \$.78; Y $.80$	X \$.67; Y $.55$	X \$.62; Y $.71$	X \$.56; Y $.61$	no validated cross-specification mapping; neither enters P_Θ; analyst selected X after inspecting the full score

Complete: applicability; each gate status; bundle status; model prediction; evidential outcome; strongest permitted conclusion; strongest forbidden inference; smallest next study.

Accessible/digital alternative: assign each expert one column in an accessible spreadsheet; freeze individual entries before revealing the shared verdict column.

Session 3 Resource. Boundary Adjudication Profiles

Frozen Rules

- Autonomy ratio A_Θ must be at least $.70$.
- Absolute endogenous capacity J_{self} must be at least $.10$.
- Candidate reliability R_C must be at least $.80$ for P_Θ membership.
- Final positive-boundary stability R_B must be at least $.80$ for a resolved positive-bearer output.
- Pairwise separability must be at least $.80$ to support multiple candidates.
- A candidate is retained only if it is role-complete and has no qualifying proper subsystem at the same grain and horizon.

All profiles are synthetic. Role-complete means V, N_1, N_2, N_3 pass under the Session 2 thresholds. For the core profiles, intervention support / positivity, consistency, causal-model adequacy, measurement reliability, regime stability, matched internal / external intervention budgets, and candidate-size normalization are stipulated to pass. Students must name these assumptions rather than infer them from a high autonomy ratio.

Student Materials

Core. Assign one profile per team. **Extension A.** Reclassify Profile D at a one-day horizon: both agents reconstruct unfinished state from the shared memory; no matched long-horizon store replacement has been run; candidate mapping across horizons is unresolved. **Extension B.** After freezing that verdict, draw one instructor-controlled audit card.

Profile	Candidate evidence	Replacement and subset evidence
A. Cardiopulmonary support, 2-second horizon	Brain B: $J_{self}=.42$, $J_{in}=.12$, $A_\Theta=.78$, $R_C=.91$, role-complete. Organism O: $J_{self}=.55$, $J_{in}=.20$, $A_\Theta=.73$, $R_C=.89$, role-complete.	B is a qualifying proper subsystem of O. Matched external perfusion replacement preserves B’s fast role profile. Selected B has $R_B=.88$.
B. Brain-computer interface, 10-second horizon	Brain B: $J_{self}=.38$, $J_{in}=.11$, $A_\Theta=.78$, $R_C=.90$, role-complete. B+I: $J_{self}=.44$, $J_{in}=.18$, $A_\Theta=.71$, $R_C=.88$, role-complete.	Generic interface replacement preserves B; B remains qualifying alone and has $R_B=.87$.
C. Organoid-controller hybrid, 30-second horizon	Tissue T: $J_{self}=.18$, $J_{in}=.09$, $A_\Theta=.67$, $R_C=.84$, fails N_3. Controller C: $J_{self}=.14$, $J_{in}=.20$, $A_\Theta=.41$, not role-complete. T+C: $J_{self}=.39$, $J_{in}=.15$, $A_\Theta=.72$, $R_C=.87$, role-complete.	Generic controller replacement fails to restore the recurrent role. Neither subsystem qualifies; selected T+C has $R_B=.84$.
D. Adaptive dyad, 1-minute horizon	A: $J_{self}=.31$, $J_{in}=.08$, $A_\Theta=.79$, $R_C=.90$, role-complete. B: $J_{self}=.29$, $J_{in}=.07$, $A_\Theta=.81$, $R_C=.89$, role-complete. A+B: $J_{self}=.63$, $J_{in}=.20$, $A_\Theta=.76$, $R_C=.86$, role-complete.	A and B are qualifying subsets, with $R_B=.88$ and $.87$. Reciprocal interruption gives separability $.87$.

Profile	Candidate evidence	Replacement and subset evidence
E. Artificial service, 100-cycle horizon	Model call M fails N_3. Agent A = model + writable state + scheduler: $J_{self}=.46$, $J_{in}=.12$, $A_\Theta=.79$, $R_C=.86$, role-complete. Full service: $J_{self}=.52$, $J_{in}=.20$, $A_\Theta=.72$, $R_C=.84$, role-complete.	A is a qualifying subsystem and has $R_B=.83$. Generic logging replacement preserves A; state-insensitive scheduler replacement fails.

Submit the observational universe, preferred candidate set, identification assumptions, intervention-budget and size-comparability checks, constituent/interface/support classification, minimality result, boundary status, and most informative next perturbation.

Accessible/digital alternative: provide one profile as a nested candidate tree and the same information as a linear list. Students may annotate either representation.

Session 4 Resource. Selective N_1 Design Sprint

Frozen Rules and Baseline

All cases are synthetic. The baseline frozen P_Θ referent has $R_C = .91$ and a separately passing cross-condition mapping; $V = .80[.75,.85]$, $N_1 = .78[.72,.84]$, $N_2 = .74[.68,.80]$, and $N_3 = .69[.62,.76]$. Use the Session 2 thresholds. A retained gate is equivalent only when its full paired-change interval is contained in the closed margin $[-.08,.08]$ and its post-intervention lower bound meets or exceeds threshold. The intervention must lower N_1 by at least $.20$.

Student Materials

Core. Select the best intervention from the table and complete a single-failure study canvas. **Extension.** Draw one complication card.

Synthetic intervention	R_C / mapping status	V	N_1	N_2	N_3	Independent anchor change
I. Cross-module phase jitter	$.90$ / pass	$.79[.74,.84]$	$.43[.38,.48]$	$.71[.66,.76]$	$.66[.60,.72]$	$-.18[-.25,-$
II. Global gain suppression	$.84$ / pass	$.52[.46,.58]$	$.40[.34,.46]$	$.36[.30,.42]$	$.32[.26,.38]$	$-.31[-.39,-$
III. Common-source removal	$.90$ / pass	$.78[.72,.84]$	$.71[.65,.77]$	$.73[.67,.79]$	$.68[.61,.75]$	$-.02[-.08,.($

Use these paired-change intervals for the retained-gate equivalence decision:

Intervention	ΔV	ΔN_2	ΔN_3
I	$[-.04,.02]$	$[-.07,.01]$	$[-.07,.01]$
II	$[-.34,-.22]$	$[-.44,-.30]$	$[-.45,-.31]$
III	$[-.05,.01]$	$[-.04,.02]$	$[-.05,.03]$

Complete this single-failure study canvas:

Field	Team entry
Exact target, candidate, regime, and horizon	
Manipulation and intended N_1 mechanism	
N_1 change and meaningful-change rule	
Frozen-referent and retained-gate equivalence checks; what follows about B^+ and what remains underdetermined	
Anchor genealogy and independence firewall	
Full-model prediction	
N_1-ablated rival prediction	
Negative control and pathway-specific rescue	

Field	Team entry
Licensed status and scope	
Result that would remain inconclusive	

Accessible/digital alternative: reveal the complication as a private LMS message. A student may submit a written selectivity audit instead of presenting the design orally.

Session 5 Resource. Causal-Routing Laboratory

Frozen N_2 Rules

All architectures are synthetic.

- A recipient qualifies only if its effect lower bound is at least $.08$, latency is no more than 180 ms, and content selectivity, route dependence, and context robustness all pass.
- Aggregate requirements: weighted coverage $C_A \geq .60$, at least three independently calibrated recipient classes, and normalized evenness $H_A \geq .80$.
- The result must remain positive after separately removing report classes and executive classes. Weights are renormalized within each reduced family.
- The class taxonomy is frozen before target-content testing.
- Availability stability must satisfy $R_A \geq \rho_A = .80$ across the frozen reasonable-specification neighborhood.
- A route-negative result requires adequate coverage of nominated redundant route families. Class count is negative only when fewer than three classes could pass even at their upper bounds; otherwise it is indeterminate.

For qualifying-class indicator a_k, point effect E_k^*, and frozen class weight w_k, calculate

$$p_k = \frac{w_k E_k^* a_k}{\sum_j w_j E_j^* a_j}, H_A = -\frac{\sum_k p_k \log p_k}{\log K_A}.$$

For $K_A < 2$, define $H_A = 0$; this prevents the entropy normalization from being undefined when no class or only one class qualifies. The supplied evenness values treat point effects as exact teaching values. A real analysis requires uncertainty for H_A as well as for each class gate.

Student Materials

Core. Assign one architecture per team. Values in brackets are 95 percent intervals. **Extension.** Compare all three and identify the smallest change that reverses each result.

Synthetic Architecture A - duplicated log bus

One hundred loggers have identical calibration repertoires and therefore count as one class.

All nominated shared-bus and redundant route families were tested with adequate coverage. $R_A = .93$.

Class	Weight	Effect	Latency	Selective	Route	Context
Logging copies	1.00	$.22[.18, .26]$	40	yes	no: shared bus common drive	yes

Synthetic Architecture B - executive-heavy router

Class	Type	Weight	Effect	Latency	Selective	Route	Context
Sensory comparison	nonexecutive	.25	$.15[.11, .1$	90	yes	yes	yes
Value updating	nonexecutive	.20	$.12[.08, .1$	115	yes	yes	yes

Class	Type	Weight	Effect	Latency	Selective	Route	Context
Memory indexing	executive	.20	$.16[.12,.2($	130	yes	yes	yes
Conflict monitoring	executive	.20	$.14[.10,.1($	145	yes	yes	yes
Report preparation	report and executive	.15	$.19[.15,.2($	160	yes	yes	yes

For the full family, $H_A \approx .993$. After executive removal, only two nonexecutive classes remain.

$R_A = .87$ across valid mappings.

Synthetic Architecture C - compact diverse circuit

Class	Type	Weight	Effect	Latency	Selective	Route	Context
Sensory comparison	nonexecutiv e	.30	$.14[.10,.1($	70	yes	yes	yes
Orienting	nonexecutiv e	.25	$.12[.09,.1($	95	yes	yes	yes
Value updating	nonexecutiv e	.25	$.13[.09,.1($	110	yes	yes	yes
Homeostati c adjustment	nonexecutiv e	.20	$.11[.08,.1($	125	yes	yes	yes

For the full and reduced families, $H_A \approx .982$.

$R_A = .91$ across valid mappings.

Submit a recipient ledger, routing sketch, full-family result, N_2^{-R}, N_2^{-E}, R_A decision, final N_2 status, and smallest missing or discriminating test.

Accessible/digital alternative: give the ledger as a filterable spreadsheet with text status columns. Provide the routing sketch as an ordered edge list for students who cannot use a visual graph.

Session 6 Resource. Temporal Intervention Studio

Frozen Timeline and Rules

The trial and all results are synthetic.

- Feedforward content encoding: 40-90 ms.
- Candidate recurrent stabilization: 110-220 ms.
- Working-memory maintenance begins: 260 ms.
- Report preparation begins: 340 ms.
- N_3 threshold: $.50$.
- Initial content fidelity must have a lower bound of at least $.70$.
- Candidate reliability must satisfy $R_C \geq .80$, and cross-condition mapping validity must separately pass.
- Retained V, N_1, N_2 gates use the Session 2 thresholds. For a selectivity claim, each full paired-change interval must be contained in the closed margin $[-.08, .08]$ and its post-intervention lower bound must meet or exceed threshold.

Baseline: content fidelity $.84[.79, .89]$, $V = .80[.75, .85]$, $N_1 = .73[.67, .79]$, $N_2 = .70[.64, .76]$, $N_3 = .68[.62, .74]$, report accuracy $.86[.80, .92]$.

Student Materials

Core. Draw the frozen timeline. Each team classifies its two assigned conditions; the class synthesizes all five only after every team has stated which causal comparison its partial evidence can and cannot identify. **Extension.** Compare recurrent System R with duration-matched feedforward System F.

Extension systems: Both R and F show observable dwell of 240 ms. Cutting R's feedback edge lowers dwell to 70 ms; pathway rescue restores 230 ms. The same nominal cut changes F by 4 ms. Shortening F's unit time constant lowers dwell to 75 ms. After small perturbation, R returns to its content-specific trajectory; F decays without return.

Submit an annotated timeline or ordered event list, the licensed N_3 test, retained-component checks, recurrence-versus-duration prediction, and one timing ambiguity.

Accessible/digital alternative: provide both a visual timeline and the ordered timeline rules plus assigned release card. Students may mark intervals in a table rather than drag events on a graphic.

Session 7 Resource. Evidence Tribunal

Frozen Four-Outcome Rule

All assessments are synthetic. The positive threshold is $\tau_+ = .70$ and the negative threshold is $\tau_- = .30$. Applicability and execution quality must pass before the score is interpreted. Credible conflict precedes thresholds.

Student Materials

Core. Ensure that all four scientific outcomes appear: with four teams assign A, B, D, and F; with six teams assign A-F. **Extension.** Use C to compare threshold uncertainty with surrogate conflict and E to compare numerical passage with failed applicability. For the six-case audit, give each variant equal target-distribution weight $1/6$. Report decisive classification coverage (positive plus negative), r_I, r_U, and whether all four outcome classes are represented.

Every variant concerns the same severe-injury patient and minimal phenomenal presence during one specified 30-second trial. A reversible visual communication trial costs 1 loss unit; missing genuine command following costs 20 loss units. On these simplified teaching losses, the trial is favored only if the decision maker's probability of genuine command following exceeds $.05$. No such probability is supplied, so an action recommendation is not uniquely determined; teams must state their assumption after fixing the scientific verdict.

Submit outcome, decisive precedence rule, cause profile emphasis, strongest licensed conclusion, forbidden inference, smallest next study, and a separate action recommendation.

Accessible/digital alternative: run the tribunal in a shared document. Advocates submit 100-word briefs; the validity officer marks admissible fields; the chair releases a written ruling.

Session 8 Resource. Diagnostic-Cell Dataset

Frozen Audit Rules

The dataset is synthetic and audits N_2.

- Every component threshold is $.60$.
- A conservative diagnostic case has $U_2 < .60$ and lower bounds for V, N_1, N_3 at or above $.60$.
- Minimum meaningful held-out risk improvement is $.05$.
- Diagnostic coverage is the sum of target-population weights for valid diagnostic cases.
- Coverage of at least $.15$ is adequate for this exercise.
- Rows with dual-use anchors, retrospective selection, or an unstable / unmappable frozen P_Θ referent are unlicensed and excluded from the confirmatory profile audit.

Student Materials

Core. Analyze Rows 1-8. **Extension.** Add Rows 9-12 and perform the full independence audit.

Intervals are shown as lower-upper. R_* is full-model held-out anchor risk; R_{-2} is risk for the N_2-ablated model. Lower risk is better. Except where the disclosed fact says otherwise, the anchor vector was frozen on a separate split and combines low-demand discrimination stability, perturbational response structure, and a no-report state marker; none shares a sensor, derived feature, or recipient-route score with the N_2 predictor.

Row	Weight	V	N_1	N_2	N_3	R_*	R_{-2}	Additional disclosed fact
1	.10	.72-.82	.70-.80	.68-.78	.66-.76	.21	.22	valid
2	.10	.20-.35	.25-.40	.22-.38	.28-.43	.18	.19	applicability and measurement quality pass
3	.08	.73-.83	.71-.81	.35-.45	.69-.79	.20	.42	valid
4	.07	.70-.80	.74-.84	.40-.50	.67-.77	.25	.43	valid
5	.10	.74-.84	.37-.47	.70-.80	.68-.78	.23	.24	applicability and measurement quality pass
6	.10	.72-.82	.71-.81	.55-.65	.69-.79	.28	.31	applicability and measurement quality pass

Row	Weight	V	N_1	N_2	N_3	R_*	R_{-2}	Additional disclosed fact
7	.10	.55-.65	.72-.82	.38-.48	.68-.78	.30	.38	applicability and measurement quality pass
8	.10	.73-.83	.72-.82	.33-.43	.36-.46	.26	.35	applicability and measurement quality pass

For the core valid diagnostic set, use the supplied synthetic bootstrap interval $\Delta_2 = R_{-2} - R_* : [.12, .28]$. Freeze diagnostic membership, coverage, and the criterion-level claim before requesting the extension rows.

Submit diagnostic membership, coverage, weighted risk interpretation, criterion/mechanism/measurement claim, status, and a one-sentence scope-qualified conclusion.

Accessible/digital alternative: use a spreadsheet with a diagnostic-membership formula already entered. Students explain the formula in words and may rely on supplied weighted results if arithmetic is not the learning target.

Session 9 Resource. Character-Bridge Break Test

Frozen Bridge Rules

All data are synthetic. Before character analysis, a licensed empirical gate $\hat{P}(S, c; \Theta_P) = 1$ is frozen for every content: the same mapped P_Θ referent is used, the inclusion-minimal positive bearer is evidentially resolved under the joint-content rule, and each content belongs to the estimated positive-bearer set. This is an evidence-relative decision, not direct access to ontic membership.

The bridge is supported only if:

1. held-out neighborhood rank accuracy is at least $.80$;
2. it improves on the best nuisance model by at least $.10$;
3. the preregistered intervention changes the predicted relation in the correct direction; and
4. observed deformation differs from predicted deformation by no more than $.20$ distance units.

Failure of a valid, interval-excluding deformation prediction refutes this bridge in the tested domain. It does not reverse the presence result.

Student Materials

Contents are red (R), orange (O), green (G), and blue (B). Fit is permitted on R, O, and G only. B is held out.

Synthetic causal distances at baseline:

	R	O	G	B
R	0	1.0	3.0	2.6
O	1.0	0	2.0	1.6
G	3.0	2.0	0	1.0
B	2.6	1.6	1.0	0

The best stimulus-only nuisance model has held-out rank accuracy $.68$.

The independent phenomenal-anchor training relations, available only for R, O, and G, are:

	R	O	G
R	0	1.0	3.0
O	1.0	0	2.0
G	3.0	2.0	0

A separate report-trained decoder gives B-to-R, B-to-O, and B-to-G distances of 2.0, 1.2, and .8. It was trained on response labels collected in the same task and may be used only as a dependence-control comparison, not as an independent phenomenal anchor.

Core. Assign Variant A or B privately. Freeze each team's bridge verdict before the variants are compared.

Submit the frozen presence status, causal-geometry source, independent anchor source, nuisance comparison, training / held-out split, deformation prediction, bridge outcome, and what remains intact if the bridge fails. Add the surviving-symmetry analysis only when the symmetry extension is assigned.

Accessible/digital alternative: supply matrices as row-wise CSV text and a verbal neighborhood list. No colored embedding is required.

Session 10 Resource. Blind Schematic Readout

Component-Law Sheet

All profiles are synthetic. Use Session 2 thresholds and the following role translations:

- stable P_{Θ} membership -> licensed profile-test referent;
- resolved inclusion-minimal B^{+} membership -> positive subject bearer;
- positive V -> realized operation;
- positive N_1 -> unity;
- positive N_2 -> system-level causal presence;
- positive N_3 -> bounded temporal presence.

Legibility requires stable decomposable semantics (D), compositional readout (C), differentiated fault localization (F), and evidential independence (E). A readable result does not validate the constitutive identity.

Student Materials

Use this common directed schematic for every case. P-candidate facts -> Referent license; V -> Profile; N1 -> Profile; N2 -> Profile; N3 -> Profile; Profile + candidate lattice -> Minimal all-pass set; Resolved minimal set -> Model prediction; and Independent anchors -> Validation comparison. A component value fills a node; the arrows encode the frozen compositional law. The schematic is a readout representation, not the physical realization.

Training Case T1: stable P_{Θ} referent; V, N_1, N_2, N_3 positive; no all-pass proper subsystem; the minimal bearer is resolved; independent anchors are held out. Expected readout: referent, operation, unity, system-level causal presence, and temporal presence are role-complete; the bearer rule issues a positive prediction, while the identity remains a hypothesis.

Training Case T2: stable P_{Θ} referent; V, N_1, N_3 positive; N_2 negative under a licensed route test. Expected readout: the system-level causal-presence role is the localized profile fault; the model issues a negative prediction for the specified referent, content, and interval, and the positive-bearer set must be recomputed.

After studying T1 and T2, Readers A and B independently classify previously unseen Profiles A-D from private reveal cards before comparing. **Extension.** Add Profile E only after the core forms are locked, then calculate agreement by exact outcome and by fault family.

For each profile record only: operational outcome; role vector; localized fault if licensed; and the step that depends on the constitutive hypothesis. After both readers finish every assigned profile, complete the following **one audit matrix for the readout trial as a whole**. Use pass, limited, fail, or not tested and cite at least one profile as evidence.

Trial-level legibility property	Decision	Evidence from the blind trial
D: decomposable semantics stayed stable across profiles		
C: the frozen compositional law generated the whole-profile outputs		
F: different licensed faults yielded differentiated readouts or interventions		
E: each supplied validation stream was independent of the component rule		

Accessible/digital alternative: use the common directed edge list and linear text profile as the schematic. Blind readers complete separate forms in private document sections; the adjudicator receives both only after submission.

Session 11 Resource. Evolving Clinical Case Conference

Student Materials

Core. This self-contained synthetic dossier is not clinical advice. Keep three targets separate: wakefulness in the observed bedside window, covert command following in the high-arousal imagery trial, and full C_0 for the specified command-related content. Wakefulness is a regime target here, not a synonym for C_0. After each controlled release, complete one target-specific row for every assessable target and request exactly one next measurement.

Compact disposition form: release stage; target and time window; candidate and regime; applicability; evidence summary; workflow state or scientific outcome; cause emphasis; strongest permitted conclusion; forbidden broader inference; one next measurement; and, only after action information is released, a separate recommendation with loss assumptions.

Use these frozen rules:

- For this teaching dossier, wakefulness is positive when sustained eye opening and sleep-wake cycling are observed in the declared bedside window; those facts are stipulated to meet the rule. This says nothing by itself about C_0.
- The narrow imagery task is positive when at least two artifact-free, cross-validated runs pass and familywise false-positive probability is below $.05$.
- Full positive C_0 classification requires a stable P_Θ referent, licensed V, N_1, N_2, N_3, a resolved inclusion-minimal positive bearer, and no unresolved credible conflict.
- A negative task result is unlicensed if sensory opportunity, arousal, task auxiliaries, or data quality fails.
- **Not yet assessed** is permitted between releases when the required test has not occurred. It is a workflow marker, not a fifth final scientific outcome.
- Scientific status is frozen before loss units or action options are considered.

Stage 1 - Rounds 1 and 2

Round 1 - context and bedside card. A patient with severe brain injury shows eye opening and sleep-wake cycles but no reliable bedside command response. Sedation has been discontinued. One examiner sees a possible visual fixation; a second does not.

Round 2 - route, boundary, and viability card. Auditory brainstem responses are present. In the later high-arousal imagery window, an independent cortical speech-discrimination check passes. A discovery-only connectivity analysis nominates a thalamocortical candidate that is stable across two grains, but no held-out minimality and separability adjudication has yet resolved a positive bearer. Physiological viability is adequate in high-arousal windows and poor in low-arousal windows.

Before requesting Stage 2, submit all Stage-1 rows, identify which targets are merely not yet assessed, and request exactly one next measurement.

After the final assigned release, submit the assessment tuple, target-specific outcome, cause emphasis, forbidden broader generalization, next study, and separate action note for every assessed window.

Accessible/digital alternative: release stages through timed LMS modules. Students may submit written role memos and do not need to portray a patient or disclose personal experience.

Session 12 Resource. Cross-Substrate Transport Jigsaw

Frozen Source Measure and Transport Rule

All dossiers are synthetic. The Perturbational Role Index (PRI) is a frozen composite of perturbational differentiation, causal spread across calibrated recipient classes, and recurrent return toward a content-specific trajectory during a 300 ms human thalamocortical trial. It estimates that source-domain operational role profile; it is not a direct measure of phenomenality. In its human source domain, PRI has a positive threshold of $.65$, sensitivity $.90$, and specificity $.88$ for that exact profile. A target-domain interpretation is licensed only if all four fields pass:

1. construct preservation;
2. causal-role preservation;
3. calibration preservation, including defensible thresholds and error rates; and
4. stable boundary and operating regime.

A favorable PRI number without all four fields is unlicensed, not negative and not positive.

Student Materials

Core. Assign one private dossier card per team. **Extension.** After core commitments, compare Dossiers B and D across a short and long horizon and specify which candidate mapping must remain stable.

Submit source construct, proposed target mapping, tolerance, four transport fields, transport license, within-theory outcome, field-level model qualifier, falsifier, next study, and separate precaution note.

Accessible/digital alternative: expert groups exchange written transport memos rather than physically regroup. Each dossier is available as an accessible linear text card.

Session 13 Resource. Versioned Rival-Prediction Laboratory

Student Materials

Use Warning

The theory cards are compact teaching abstractions, not substitutes for primary texts. The assigned source governs whenever a card and source differ. Every case profile is synthetic. Before final submission, the source auditor must attach one section or page citation that supports the rival prediction or mark it unresolved.

Use these frozen source assignments and locators. If pagination differs, locate the named section and record the edition used.

Card	Frozen source	Required passage
1	Stilwell (2026), *C0 as N** Version 2.0	Sections 2, 4-12, 15, and 18
2	Albantakis et al. (2023), IIT 4.0	sections “IIT 4.0: postulates and principles” and “Exclusion postulate”
3	Mashour et al. (2020), GNWT review	pp. 780-789 on global availability, ignition, and workspace dynamics
4	Lamme (2006), recurrent-processing account	pp. 494-498 on local recurrence and later access
5	Lau and Rosenthal (2011), higher-order account	pp. 365-369 on higher-order representation and conscious awareness
6	Friston et al. (2021), active-inference framework	pp. 211-221 on Markov blankets and self-organizing dynamics; mark any consciousness criterion not stated there as unresolved
7	Graziano and Webb (2015), attention-schema theory	sections “Basic mechanism” and “The attention schema and awareness”
8	Thompson and Varela (2001), radical embodiment	pp. 418-423 on organismic regulation and brain-body dynamics
9	Frankish (2016), illusionism	pp. 11-20 on the revisionary target and explanatory project
10	Goff (2019), panpsychist fundamentalism	Introduction and Chapter 2; identify whether the source issues a matched case-level prediction

Core Version Cards

Card	Exact teaching version	Primary target	Bearer and central commitment	Report and time	Main caution
1	Cø / N* as taught in this manual	minimal phenomenal presence	independently bounded viable candidate jointly realizes integration, diverse causal availability, and bounded recurrence	report not constitutive; recurrence required	identity remains a hypothesis
2	IIT 4.0 teaching card, Albantakis et al. (2023)	phenomenal existence and structure	intrinsic cause-effect power of a candidate complex; irreducibility and exclusion are central	report not constitutive; temporal realization depends on system update	do not reduce IIT to generic integration magnitude
3	GNWT review version, Mashour et al. (2020)	conscious access or conscious processing	global availability through workspace-like ignition and broadcasting to multiple processors	report is common evidence, not always a defining output; sustained access dynamics matter	exact prefrontal and report commitments vary across GNWT formulations
4	Recurrent-processing version, Lamme (2006)	conscious perceptual content	recurrent processing in sensory systems is central	report and later access are separable; local feedback can suffice	distinguish recurrence from mere lateness
5	Higher-order representation teaching card, assigned Lau and Rosenthal (2011) source	conscious state or awareness of a first-order state	a suitable higher-order representation of the first-order state	report is evidence, not necessarily constitution; temporal details version-dependent	specify whether the theory targets minimal presence or reflective awareness

Extension Version Cards

Card	Teaching version	Primary pressure	Comparison caution
6	Predictive or active-inference account, assigned Friston et al. (2021) source	precision-weighted hierarchical inference and self-organizing regulation	may be an implementation framework rather than a unique consciousness criterion
7	Attention-schema account, assigned Graziano and Webb (2015) source	an internal model of attention supports control and consciousness attribution	often targets self-model, report, or attribution rather than minimal presence alone
8	Embodied or enactive account, assigned Thompson and Varela (2001) source	organismic regulation, action, affect, and world-involving dynamics	state whether embodiment is constitutive, enabling, or explanatory background

Card	Teaching version	Primary pressure	Comparison caution
9	Illusionist account, assigned Frankish (2016) source	explain judgments and dispositions to report phenomenality without positing phenomenal properties as C0 treats them	target may be revisionary rather than a case-level neural rival
10	Fundamentalist or panpsychist account, assigned Goff (2019) source	consciousness is fundamental and organizational questions concern combination or structure	often does not issue the same case-level prediction without a combination theory

Synthetic Diagnostic Profiles

Core. Assign one profile and its named core rival. **Extension.** Add one extension rival only when its assigned source yields a genuine matched-target prediction.

Every profile concerns one frozen candidate S, content c, regime, and interval. In A-D, S is a stable causally admissible P_{Θ} referent and the stated licensed component failures block its positive-bearer membership; conclusions concern that specified referent, not every possible bearer elsewhere in the system. Profile E supplies the additional minimality and resolution facts needed for a positive-bearer verdict. Receive only the profile assigned to your team until predictions are committed.

Submit the exact source version and passage, matched target, ordinary agreement case, diagnostic profile, prospective predictions, decisive evidence, and revision commitment for both accounts.

Accessible/digital alternative: version cards are searchable text. Advocates may submit source-verified written predictions before discussion; the comparison chair works from those records.

Session 14 Resource. Adversarial Preregistration and Result Tribunal

Student Materials

Frozen Practice Specification

All practice materials are synthetic. Use the compact capstone template below. If a student team has its own capstone, its frozen rules replace these practice values.

- Lower risk is better. Define model-risk difference as $\Delta_R = R_{rival} - R_{full}$ and the minimum meaningful difference as $.05$.
- Adequate diagnostic coverage: at least $.20$ of the target distribution.
- **Full-model criterion support:** every license passes, coverage is at least $.20$, and $LB(\Delta_R) > .05$.
- **Equivalence or redundancy:** every applicable license passes, coverage is at least $.20$, the complete interval for Δ_R is contained in the closed margin $[-.05, .05]$, retained-component and sensitivity checks pass, model flexibility is matched, transport is matched when a cross-domain claim is made, and no compensatory retuning occurs.
- **Rival-favoring:** every applicable license passes, coverage is at least $.20$ for a population-scope conclusion, and $UB(\Delta_R) < -.05$.
- **Indeterminate:** the interval crosses or touches a decision boundary, credible anchors conflict, or coverage is inadequate for the declared population scope. In particular, equality of the relevant bound at $+.05$ or $-.05$ is indeterminate under these strict meaningful-effect rules. A low-coverage effect may be described locally but cannot support a broad full-model or rival-favoring conclusion.
- **Unlicensed:** referent/boundary, applicability, quality, independence, intervention selectivity, or, when a cross-domain claim is made, transport fails before risk adjudication.
- Retained-component equivalence requires each complete paired-change interval to be contained in the closed margin $[-.08, .08]$; a point estimate is insufficient.
- Boundary stability passes when its lower bound meets or exceeds $.80$. Any card that supplies only a point value stipulates it as an exact teaching value.
- Credible anchor conflict: nonoverlapping intervals and point-estimate difference greater than $.15$.
- Credible anchor conflict has precedence over averaging and over the risk-difference rules.
- When a cross-domain claim is made, transport requires construct, causal-role, calibration, and boundary preservation. A source-domain audit is not unlicensed merely because transport was not attempted.

Synthetic Starter Protocol

Target: minimal phenomenal presence of one auditory content from 120-240 ms in a frozen recurrent preparation. Full model: V, N_1, N_2, N_3 are jointly required. Rival: a duration-matched model omits recurrence topology while retaining V, N_1, N_2. Diagnostic contrast: selective return-path interruption with content fidelity and retained components equivalent. Primary anchor: held-out perturbational discrimination vector sharing no sensor, feature, or route with the N_3 measure. Negative control: late report-path interruption. Rescue: pathway-specific feedback restoration.

Core studio task: Complete only these minimum fields before red-team review: target and exclusion; exact rival predictions; diagnostic cell; candidate, regime, and interval; intervention and retained-gate checks; primary anchor and prohibited dual use; one negative control; and the outcome-to-revision table.

Compact capstone field	Frozen team entry
Exact target, exclusions, candidate, regime, interval	
Exact full-model and rival versions	
Ordinary agreement region and diagnostic contrast	
Candidate proposal; frozen P_{Θ} stability; post-result B^{+} recomputation	
Manipulation, retained-component equivalence, rescue	
Anchor vector and independence firewall	
Component-audit disposition: full-model supported, equivalent/redundant, rival-favoring, indeterminate, or unlicensed	
Separate presence-classification rule, if the study also classifies a candidate: positive, negative, indeterminate, or unlicensed	
Result-to-revision commitments for both accounts	
Feasibility, ethics, data, and stopping rule	

Extension: add a causal diagram, variable dictionary, simulated-data recovery plan, and team role / dissent charter.

After the team freezes its rules, the instructor assigns one result card without warning. The tribunal submits three sentences: **result**, **inference licensed**, and **next commitment**.

Accessible/digital alternative: publish question domains in advance, allow two quiet minutes before answers, and permit a short written supplement. The result tribunal can run asynchronously in a disposition log.